

Trade Tariffs and Exchange Rates in a Multilateral World

Streamlined model derivations

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1 Introduction

How does the conventional two-country result, that an import tariff appreciates the tariffing country's currency, change in a multilateral setting? The answers derived below:

- In the symmetric benchmark, a unilateral tariff always appreciates A against the tariff target B .
- With a third country, A can *depreciate* against an untreated country C when cross-origin substitution is sufficiently strong ($\rho > 3\rho_1^*$). The bilateral ambiguity reflects reallocation of expenditure across foreign suppliers.
- Under symmetry, this reallocation cancels from the numerator of A 's effective exchange-rate response, so the conventional appreciation result survives in aggregate: the NEER appreciates for every ρ .
- With N countries, diversion toward each individual bystander is diluted, raising the bilateral reversal threshold to $N\rho_1^*$; the NEER result is unchanged.
- Tariff configuration matters: a broad tariff deactivates the reallocation margin, while reciprocal retaliation cancels the belligerents' bilateral component and can turn the reallocation margin decisive even for effective rates ($\rho > \rho_1^*$).
- Asymmetry qualifies the exact NEER result: it is preserved to leading order within a structured asymmetric family, and can reverse when the target is disproportionately small.
- The canonical Viner–Mundell PTA result emerges when the margins opposing foreign-supplier diversion are removed.

The results sit at four levels of generality. Any differentiable multilateral equilibrium separates the tariff's trade-balance disturbance F from the exchange-rate adjustment mechanism J . The broad mechanism is the distinction between aggregate home-versus-foreign adjustment and reallocation across foreign suppliers. The nested CES demand system represents these margins with two constant elasticities, η and ρ . Symmetry then delivers exact linear thresholds and the exact NEER cancellation. The exact formulas belong to the bottom layer; the mechanism belongs to the layers above it. Every closed form is verified numerically in the replication package.

2 Economic Environment

2.1 Production

Each country $i \in \{1, \dots, N\}$ produces a tradable T_i (only i produces it, all countries consume it) and a non-tradable N_i , from labor alone:

$$Y_{T_i} = A_{T_i} L_{T_i}, \quad Y_{N_i} = A_{N_i} L_{N_i}, \quad L_{T_i} + L_{N_i} = L_i. \quad (1)$$

Perfect competition and the normalization $P_{T_i} = 1$ (producer price in i 's own currency) give

$$w_i = A_{T_i}, \quad P_{N_i} = \frac{w_i}{A_{N_i}} = \frac{A_{T_i}}{A_{N_i}}. \quad (2)$$

Producer prices are unit labor costs and wages are constant in own currency, so all relative adjustment runs through exchange rates.

2.2 Preferences

$$C_i = C_{T_i}^{\alpha_T} C_{N_i}^{1-\alpha_T} \quad (\text{Cobb-Douglas outer, tradable share } \alpha_T) \quad (3)$$

$$C_{T_i} = \left[\alpha_D^{1/\eta} C_{T_i}^{\frac{\eta-1}{\eta}} + (1 - \alpha_D)^{1/\eta} M_i^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}} \quad (\text{home vs. imports, elasticity } \eta, \text{ home weight } \alpha_D) \quad (4)$$

$$M_i = \left[\sum_{j \neq i} \beta_{ij}^{1/\rho} C_{T_j}^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}} \quad (\text{across origins, elasticity } \rho, \text{ weights } \sum_{j \neq i} \beta_{ij} = 1) \quad (5)$$

with the share-linear CES convention. η governs the **home-versus-foreign margin**: how much expenditure moves between domestic and imported tradables. ρ governs the **foreign-supplier reallocation margin**: conditional on importing, how readily expenditure moves across foreign origins. At $N = 2$ there is one foreign supplier and the second margin is inactive; it becomes economically active at $N \geq 3$. The distinction is more general than the CES specification: any demand system in which total import expenditure and its allocation across origins adjust separately contains analogous margins. Nested CES isolates them with two constant elasticities and thereby yields closed-form thresholds; the mechanism is general, the parameterization and exact theorems are model-specific.

The symmetric benchmark imposes equal origin weights $\beta_{ij} = 1/(N - 1)$, common preference parameters $(\alpha_T, \alpha_D, \eta, \rho)$, and equal endowments and productivities, so countries have equal size at the free-trade baseline. Home and foreign tradables remain asymmetric through α_D ; full symmetry across all tradables ($\rho = \eta$, $\alpha_D = 1/N$) is not imposed. The single-elasticity CES is $\rho = \eta$; Cobb-Douglas within tradables is $\eta = 1$, which fixes the elasticity, not the weight α_D .

2.3 Policy, income, and trade balance

Tariffs $\tau_{ij} \geq 0$ are ad valorem, levied by i on imports from j ; the consumer price of T_j in i is $p_{ij} = (1 + \tau_{ij}) e_{ij}$ in i 's currency, with $\tau_{ii} = 0$ as a normalization so that $p_{ii} = 1$. CES price indices in i 's currency:

$$P_{M_i} = \left[\sum_{j \neq i} \beta_{ij} p_{ij}^{1-\rho} \right]^{\frac{1}{1-\rho}}, \quad P_{T_i} = \left[\alpha_D p_{ii}^{1-\eta} + (1 - \alpha_D) P_{M_i}^{1-\eta} \right]^{\frac{1}{1-\eta}}, \quad (6)$$

$$P_i = \kappa_i P_{T_i}^{\alpha_T} P_{N_i}^{1-\alpha_T} \quad (\kappa_i \text{ collects share constants}). \quad (7)$$

Expenditure shares of income (rows sum to α_T over tradables):

$$s_{ii} = \alpha_T \alpha_D \left(\frac{p_{ii}}{P_{T_i}} \right)^{1-\eta}, \quad (8)$$

$$s_{ij} = \alpha_T (1 - \alpha_D) \left(\frac{P_{M_i}}{P_{T_i}} \right)^{1-\eta} \beta_{ij} \left(\frac{p_{ij}}{P_{M_i}} \right)^{1-\rho}, \quad j \neq i. \quad (9)$$

Shares are tariff-inclusive: $s_{ij} I_i$ is spending at consumer prices; $s_{ij} I_i / (1 + \tau_{ij})$ is the border value. Revenue is rebated lump-sum,

$$I_i = w_i L_i + \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij} I_i \implies I_i = \frac{w_i L_i}{1 - \sum_{j \neq i} \frac{\tau_{ij}}{1 + \tau_{ij}} s_{ij}}, \quad (10)$$

a closed form since the shares depend only on prices. Trade balances are measured at producer prices in a common currency (the tariff wedge is a domestic transfer):

$$\text{TB}_i = \underbrace{\sum_{k \neq i} f_{ki}}_{\text{exports}} - \underbrace{\sum_{j \neq i} f_{ij}}_{\text{imports}}, \quad f_{ij} \equiv \frac{s_{ij} I_i}{1 + \tau_{ij}}. \quad (11)$$

By Walras' law $\sum_i \text{TB}_i = 0$, so only $N - 1$ conditions are independent, matching the $N - 1$ free exchange rates.

Assumption 1 (Closure). *There are no internationally traded assets: $\text{TB}_i = 0$ for all i , every period.*

Given $\{\tau_{ij}\}$, the rates $\{e_{Aj}\}_{j \neq A}$ solve $\text{TB}_i = 0$ for $i \neq A$. A tariff first changes trade flows at the existing exchange rates; exchange rates then move until the resulting imbalances disappear.

2.4 Exchange-rate concepts

e_{ij} = units of i 's currency per unit of j 's currency, $e_{ii} = 1$, $e_{ij} = e_{ik}e_{kj}$. Sign convention: **a rise in e_{ij} is a depreciation of i against j** . Perturbations: $\nu_j \equiv d \log e_{Aj}$ with $\nu_A = 0$.

$$\omega_{ij} \equiv \frac{e_{ij}w_j}{w_i} \propto e_{ij} \quad (\text{relative wage, common currency}) \quad (12)$$

$$\text{ToT}_{ij} \equiv \frac{P_{T_i}}{e_{ij}P_{T_j}} = \frac{1}{e_{ij}} \quad (\text{bilateral terms of trade, pre-tariff prices}) \quad (13)$$

$$q_{ij} \equiv \frac{e_{ij}P_j}{P_i} \quad (\text{bilateral real exchange rate}) \quad (14)$$

$$d\nu_i^E \equiv \sum_{j \neq i} w_{ij} d \log e_{ij}, \quad w_{ij} = \frac{f_{ij} + f_{ji}}{\sum_{k \neq i} (f_{ik} + f_{ki})} \quad (\text{NEER, trade weights}) \quad (15)$$

$$dq_i^E \equiv \sum_{j \neq i} w_{ij} d \log q_{ij} \quad (\text{REER, same weights}) \quad (16)$$

with f_{ij} from (11); the currency choice cancels from the weights, and $w_{ij} = 1/(N-1)$ in the symmetric model. Effective rates are depreciation-positive. Because producer prices are unit labor costs and demands depend on (w_j, e_{ij}) only through $e_{ij}w_j$,

$$d \log e_{ij} = d \log \omega_{ij} = -d \log \text{ToT}_{ij} : \quad (17)$$

every exchange-rate result below can equivalently be read as a relative-wage or terms-of-trade result.

3 Two Countries: Conventional Wisdom

3.1 The equilibrium response

A imposes a small tariff $d\tau$ on B at the symmetric balanced free-trade baseline (incomes normalized to one). Write $m \equiv \alpha_T(1 - \alpha_D)$ for the baseline import share, so $s_{AB} = s_{BA} = m$. With $\nu_B = d \log e_{AB}$, own-currency price changes are $d \log p_{AB} = \nu_B + d\tau$ and $d \log p_{BA} = -\nu_B$; with a single origin these equal the import-index changes, and $d \log P_{T_i} = (1 - \alpha_D) d \log P_{M_i}$. Shares respond only on the home-versus-import margin,

$$d \log s_{AB} = (1 - \eta) \alpha_D (\nu_B + d\tau), \quad d \log s_{BA} = -(1 - \eta) \alpha_D \nu_B, \quad (18)$$

incomes move by $d \log I_A = m d\tau$ (rebated revenue) and $d \log I_B = \nu_B$ (common currency), and the border value of imports loses the wedge $d\tau$. Differentiating TB_A in (11) flow by flow and collecting terms:

$$d \text{TB}_A = \underbrace{m D_2 \nu_B}_{\text{adjustment}} + \underbrace{m \rho_1^* d\tau}_{\text{impact disturbance}} = 0, \quad D_2 \equiv 2\alpha_D(\eta - 1) + 1, \quad (19)$$

so that

$$\boxed{\left. \frac{d \log e_{AB}}{d\tau} \right|_{\tau=0}} = -\frac{\rho_1^*}{D_2}, \quad \rho_1^* = 1 + \alpha_D(\eta - 1) - \alpha_T(1 - \alpha_D) = \underbrace{\alpha_D \eta}_{\text{home switching}} + \underbrace{(1 - \alpha_D)(1 - \alpha_T)}_{\text{expenditure absorption}} > 0. \quad (20)$$

Setting $\nu_B = 0$ in (19): at unchanged exchange rates the tariff improves A 's balance by $m\rho_1^*d\tau > 0$, through switching toward the home tradable and absorption by the non-tradable margin. The coefficient mD_2 is the rate at which A 's depreciation improves its balance. The notation ρ_1^* anticipates its later role as the per-partner unit of every multilateral threshold.

3.2 Equivalence with the textbook Marshall–Lerner condition

In reduced form, let export volume be $X(p^*)$ with $p^* = 1/e$ and import volume $M(p)$ with $p = (1 + \tau)e$, with elasticities $\varepsilon_X \equiv -X'p^*/X \geq 0$, $\varepsilon_M \equiv -M'p/M \geq 0$, and

$$\text{TB}_A(e, \tau) = X(1/e) - eM((1 + \tau)e). \quad (21)$$

At $\tau = 0$ and balanced trade ($X = eM$),

$$\frac{\partial \text{TB}_A}{\partial e} = \underbrace{\frac{\varepsilon_X X}{e}}_{\text{export volume}} + \underbrace{\frac{\varepsilon_M M}{\text{import volume}}}_{\text{import volume}} - \underbrace{\frac{M}{\text{valuation}}}_{\text{valuation}} = M(\varepsilon_X + \varepsilon_M - 1), \quad \frac{\partial \text{TB}_A}{\partial \tau} = \frac{\varepsilon_M}{1 + \tau} eM > 0, \quad (22)$$

and the implicit function theorem gives

$$\frac{de}{d\tau} = -\frac{\partial \text{TB}_A / \partial \tau}{\partial \text{TB}_A / \partial e}, \quad \text{sign } \frac{de}{d\tau} = -\text{sign}(\varepsilon_X + \varepsilon_M - 1). \quad (23)$$

In the nested model the elasticity condition is not free-standing: comparing (22) with (19) per unit of import value,

$$\varepsilon_X + \varepsilon_M - 1 = D_2, \quad \varepsilon_X = \varepsilon_M = \alpha_D(\eta - 1) + 1. \quad (24)$$

Result 3.1 (Conventional wisdom, $N = 2$). *Under the Marshall–Lerner condition $\varepsilon_X + \varepsilon_M > 1$, equivalently $D_2 > 0$ (automatic for $\eta \geq 1$; failing only for $\eta < 1 - 1/(2\alpha_D)$), a unilateral import tariff appreciates the tariffing country's currency: $d \log e_{AB}/d\tau = -\rho_1^*/D_2 < 0$ since $\rho_1^* > 0$.*

At fixed exchange rates the tariff creates a surplus for A ; under Marshall–Lerner an appreciation eliminates it. The disturbance signs the numerator, the adjustment mechanism signs the denominator, and at $N = 2$ both are unambiguous.

3.3 Exact Cobb–Douglas benchmark

At $\eta = 1$ shares are constant, $I_A = w_A L_A / [1 - \frac{\tau}{1+\tau} m]$ from (10), and $TB_A = 0$ reads $e m w_B L_B = m I_A / (1 + \tau)$:

$$\boxed{e(\tau) = \frac{w_A L_A}{w_B L_B} \cdot \frac{1}{1 + (1 - m)\tau}}, \quad \left. \frac{d \log e}{d\tau} \right|_{\tau=0} = -(1 - m) = -\left. \frac{\rho_1^*}{D_2} \right|_{\eta=1}, \quad (25)$$

globally decreasing in τ : the linearized sign is exact at all tariff levels in this case. ($\eta = 1$ pins shares at the weights but leaves α_D free.)

4 Three Countries: Bilateral Ambiguity

4.1 The system and its solution

Add an untreated country C , symmetric with B ; A tariffs only B ($d\tau = dt_{AB}$). Expenditure displaced from B can now be redirected toward C , activating the foreign-supplier reallocation margin. Baseline shares are $s_{ii} = \alpha_T \alpha_D$, $s_{ij} = m/2$; own-currency price changes $dp_{ij} = \nu_j - \nu_i + dt_{ij}$ give

$$d \log P_{MA} = \frac{1}{2}(\nu_B + \nu_C + d\tau), \quad d \log P_{MB} = \frac{1}{2}\nu_C - \nu_B, \quad d \log P_{MC} = \frac{1}{2}\nu_B - \nu_C, \quad (26)$$

with $d \log I_A = \frac{m}{2} d\tau$, $d \log I_B = \nu_B$, $d \log I_C = \nu_C$. Shares now respond on both margins, as in (36) below with elasticities ρ (across origins) and η (home versus imports). Substituting into the differentiated trade balances for $i = B, C$ and collecting terms in $(\nu_B, \nu_C, d\tau)$:

$$J \begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = -F d\tau, \quad J = -\frac{mD}{4} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad F = -\frac{m}{4} \begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix}, \quad (27)$$

with $D \equiv D_3 = 3\alpha_D(\eta - 1) + \rho + 1$; the elasticities enter J only through the scalar D . The impact effects at unchanged exchange rates are

$$F_B = -\frac{m}{4}(\rho + \rho_1^*) < 0, \quad F_C = \frac{m}{4}(\rho - \rho_1^*) : \quad (28)$$

B necessarily loses export demand; C gains diverted expenditure (increasing in ρ) but loses sales to a poorer B and to A 's home switching, with net gain iff $\rho > \rho_1^*$. Inverting with $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$:

$$\begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = (-J)^{-1} F d\tau = -\frac{1}{3D} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} \rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau = \frac{1}{3D} \begin{pmatrix} -(\rho + 3\rho_1^*) \\ \rho - 3\rho_1^* \end{pmatrix} d\tau. \quad (29)$$

Result 4.1 (Threshold). *At the symmetric point, with $D > 0$ (automatic for $\eta \geq 1$, $\rho > 0$),*

$$\frac{d \log e_{AB}}{d\tau} = -\frac{\rho + 3\rho_1^*}{3D} < 0, \quad \boxed{\frac{d \log e_{AC}}{d\tau} = \frac{\rho - 3\rho_1^*}{3D}} \quad \frac{d \log e_{BC}}{d\tau} = \frac{2\rho}{3D} > 0. \quad (30)$$

A appreciates against B and B depreciates against C for all parameters; A depreciates against the untariffed C iff $\rho > \rho_3^ \equiv 3\rho_1^*$.*

The tariff shifts expenditure away from *B*; sufficiently strong cross-origin substitution redirects enough of it toward *C* that *C* appreciates not only against *B* but against *A*. Note the gap between the impact and equilibrium thresholds. From (29),

$$\frac{d\nu_C}{d\tau} \propto F_B + 2F_C \propto \rho - 3\rho_1^* : \quad (31)$$

$F_C > 0$ at $\rho > \rho_1^*$, but *C*'s equilibrium rate mixes its own gain with *B*'s loss, so reversal requires $\rho > 3\rho_1^*$. More generally, *a bystander's own favorable diversion shock need not determine its equilibrium bilateral rate, because that rate responds to the entire disturbance vector through $(-J)^{-1}F$.*

4.2 Corollary: impossibility under a single elasticity

Impose $\rho = \eta = \sigma$. Then

$$\rho_3^* - \sigma = \sigma(3\alpha_D - 1) + 3(1 - \alpha_D)(1 - \alpha_T) > 0 \quad \text{whenever } \alpha_D \geq 1/3 : \quad (32)$$

with one elasticity the reversal against *C* is unreachable on the home-biased region, for any α_T . At the equal-weights illustration $\alpha_D = 1/3$, $\alpha_T = 3/4$ (the paper's original calibration, not required): $\rho_3^* - \sigma = \frac{1}{2}$, $D = 2\sigma$, and $d \log e_{AC}/d\tau = -1/(12\sigma) \rightarrow 0^-$.

5 Disturbance and Adjustment in General Equilibrium

The three-country case suggests the general decomposition

$$\text{tariff} \longrightarrow \text{trade-balance disturbances } F \longrightarrow \text{exchange-rate adjustment } (-J)^{-1}F.$$

The decomposition is the most general layer of the analysis: any differentiable multilateral trade environment with $N - 1$ independent trade-balance conditions admits a local representation $J d\nu + G d\tau = 0$. The nested CES structure enters by supplying the explicit entries of J and G derived below, and primitive conditions for their signs.

5.1 General linearization

Evaluate at $\tau_{ij} = 0$ with balanced trade and normalized incomes. Baseline objects: home share $h_i = \alpha_D$ (a country-specific h_i carries through unchanged), within-import shares $b_{ij} = \beta_{ij}$, income shares $s_{ij} = m b_{ij}$, flow values $f_{ij} = s_{ij}I_i$, with $\sum_k f_{ki} = \sum_j f_{ij}$. Perturb by tariffs dt_{ij} :

$$dp_{ij} = \nu_j - \nu_i + dt_{ij} \quad (dp_{ii} = 0), \quad (33)$$

$$d \log P_{M_i} = \sum_{j \neq i} b_{ij} dp_{ij}, \quad d \log P_{T_i} = (1 - \alpha_D) d \log P_{M_i}, \quad (34)$$

$$d \log s_{ii} = -(1 - \eta)(1 - \alpha_D) d \log P_{M_i}, \quad (35)$$

$$d \log s_{ij} = \underbrace{(1 - \rho)(dp_{ij} - d \log P_{M_i})}_{\text{across origins}} + \underbrace{(1 - \eta) \alpha_D d \log P_{M_i}}_{\text{home vs. imports}}, \quad (36)$$

$$d \log I_i = \nu_i + \sum_{j \neq i} s_{ij} dt_{ij}, \quad (37)$$

$$d \text{TB}_i = \sum_{k \neq i} f_{ki} (d \log s_{ki} + d \log I_k - dt_{ki}) - \sum_{j \neq i} f_{ij} (d \log s_{ij} + d \log I_i - dt_{ij}). \quad (38)$$

Stack $d\nu \equiv (\nu_j)_{j \neq A}$ and $d\tau \equiv (dt_{jk})_{j \neq k}$. Reading coefficients off (38), the conditions for $i \neq A$ are

$$J d\nu + G d\tau = 0, \quad J_{il} \equiv \frac{\partial d \text{TB}_i}{\partial \nu_l}, \quad G_{i,(jk)} \equiv \frac{\partial d \text{TB}_i}{\partial dt_{jk}} \Big|_{\nu \text{ fixed}}. \quad (39)$$

A policy experiment is a direction $a = (a_{jk})$ scaled by $d\tau$ (unilateral tariff: $a_{AB} = 1$; war: $a_{AB} = a_{BA} = 1$), with impact vector

$$F \equiv Ga, \quad F_i = \frac{\partial d \text{TB}_i}{\partial d\tau} \Big|_{\nu=0}, \quad \implies \quad J d\nu = -F d\tau, \quad \frac{d\nu}{d\tau} = (-J)^{-1} F = \frac{\text{adj}(-J) F}{\det(-J)}. \quad (40)$$

F is the trade-balance disturbance at unchanged exchange rates; J is the adjustment mechanism. Different configurations are different directions a applied to the same G (the superposition used in Section 8). At $N = 3$, (40) is (27); at $N = 2$ it is (19).

5.2 The multilateral Marshall–Lerner condition and the sign rule

Write $\tilde{J}$ for the full $N \times N$ Jacobian before dropping country A . Two identities of the model:

$$\sum_l \tilde{J}_{il} = 0 \quad \forall i \quad (\text{homogeneity}), \quad \sum_i \tilde{J}_{il} = 0 \quad \forall l \quad (\text{Walras' law}). \quad (41)$$

Assumption 2 (Gross substitutability). $\tilde{J}_{il} \geq 0$ for all $l \neq i$: a depreciation of any country does not worsen any other country's trade balance. In the symmetric systems of Sections 4 and 6, every off-diagonal entry is a positive multiple of D_N , so the assumption there is $D_N > 0$, which holds for $\eta \geq 1$.

Homogeneity plus Assumption 2 give the diagonal, and the reduced J inherits a dominant diagonal:

$$\tilde{J}_{ii} = - \sum_{l \neq i} \tilde{J}_{il} \leq 0, \quad |J_{ii}| = \sum_{l \neq i, A} J_{il} + \tilde{J}_{iA} \geq \sum_{l \neq i, A} J_{il}, \quad (42)$$

the first being the bilateral Marshall–Lerner conditions (one per country), the second strict in every row with $\tilde{J}_{iA} > 0$. Dominance is therefore derived, not assumed. By the standard characterization of irreducibly diagonally dominant Z-matrices:

Result 5.1 (Multilateral Marshall–Lerner condition). *Under Assumption 2, with J irreducible and*

$\tilde{J}_{iA} > 0$ for some i , $-J$ is a nonsingular M -matrix:

$$\det(-J) > 0, \quad (-J)^{-1} \geq 0 \text{ elementwise}, \quad \operatorname{Re} \lambda(J) < 0, \quad (43)$$

three equivalent statements referred to jointly as the multilateral Marshall–Lerner condition.

Gross substitutability is sufficient, not necessary; everything below uses only (43). The eigenvalue face is local stability of the tatonnement $\dot{\nu}_i = k_i \text{TB}_i$ for any speeds $k_i > 0$ (a surplus country appreciates), so stability is not a separate requirement. At $N = 2$, (43) collapses to $J < 0$, i.e. $D_2 > 0$, whose elasticity form is (24).

Result 5.2 (Sign rule). *Under (43),*

$$\operatorname{sign}\left(\frac{d\nu_i}{d\tau}\right) = \operatorname{sign}\left(\sum_j \omega_{ij} F_j\right), \quad \omega_{ij} \equiv [(-J)^{-1}]_{ij} \geq 0 : \quad (44)$$

every response is a nonnegative combination of the impact effects, so opposite-signed bilateral responses require a mixed-sign disturbance vector.

This locates the three-country ambiguity: $F_B < 0$ always, F_C turns positive through diversion, and the bilateral ambiguity comes from F , never from a failure of the adjustment mechanism. Since F is linear in ρ and $\operatorname{adj}(-J)$ has degree $N - 2$, each weighted sum is a degree- $(N - 1)$ polynomial in ρ : one relevant root at $N = 3$, possibly several sign changes at $N \geq 4$.

6 Aggregation: The NEER

If A appreciates against B but can depreciate against C , what happens to the overall value of A 's currency? Decompose (29):

$$\underbrace{\frac{\nu_B + \nu_C}{2}}_{\text{NEER of } A} = -\frac{\rho_1^*}{D} d\tau, \quad \underbrace{\nu_B - \nu_C}_{\text{relative}} = -\frac{2\rho}{3D} d\tau. \quad (45)$$

Result 6.1 (NEER preservation). *With symmetric weights, $d\nu_A^E/d\tau = -\rho_1^*/D < 0$ for every ρ satisfying (43): a unilateral tariff appreciates the tariffing country's effective exchange rate even when $\rho > \rho_3^*$ reverses the bilateral rate against C .*

ρ governs the relative allocation of adjustment across foreign currencies: its direct terms cancel in the aggregate numerator, although it remains in D and therefore affects the magnitude of the NEER response. The two-country appreciation result is thus best interpreted as an aggregate home-versus-foreign result that is also bilateral only because there is one foreign country. The exact cancellation is a property of the symmetric nested-CES environment (Section 10 quantifies its robustness); the distinction between aggregate adjustment and foreign-supplier reallocation is more general.

Empirical location. Given (43), the signs of e_{AB} , e_{BC} , and the NEER are parameter-free; only the bystander rate is ambiguous. Locating its threshold empirically requires care: ρ is substitution across foreign origins conditional on the import composite, and estimates labeled “trade elasticities” generally measure different objects. Qualitatively, upper-nest (home-versus-import) estimates are low and lower-nest (cross-origin) estimates several times larger, so the benchmark $\rho_3^* = 3.96$ ($\alpha_D = 0.8$, $\alpha_T = 0.4$, $\eta = 1.5$) is neither clearly above nor below plausible ρ : neither sign of e_{AC} can be ruled out. The war threshold $\rho_1^* = 1.32$ (Section 8) is less sensitive to this mapping, so the war predictions are comparatively robust.

7 General Symmetric N : Dilution

A tariffs B ; the $N - 2$ bystanders are identical with common ν_C and weights $b_{ij} = 1/(N - 1)$. From (33)–(34):

$$d \log P_{M_A} = \frac{\nu_B + d\tau + (N - 2)\nu_C}{N - 1}, \quad d \log P_{M_B} = \frac{(N - 2)\nu_C}{N - 1} - \nu_B, \quad d \log P_{M_C} = \frac{\nu_B - 2\nu_C}{N - 1}, \quad (46)$$

with $d \log I_A = \frac{m}{N-1} d\tau$, $d \log I_B = \nu_B$, $d \log I_C = \nu_C$. Substituting into (38) for B and a representative bystander and collecting terms (rows scaled by $(N - 1)/m$; verified symbolically):

$$D_N \begin{pmatrix} -(N - 1) & N - 2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} \nu_B \\ \nu_C \end{pmatrix} = \begin{pmatrix} (N - 2)\rho + \rho_1^* \\ \rho_1^* - \rho \end{pmatrix} d\tau, \quad D_N = N\alpha_D(\eta - 1) + (N - 2)\rho + 1, \quad (47)$$

where the elasticities enter the adjustment matrix only through the scalar D_N and the counting matrix has determinant N ; inverting:

$$\frac{d \log e_{AB}}{d\tau} = -\frac{N\rho_1^* + (N - 2)\rho}{N D_N} < 0, \quad \frac{d \log e_{AC}}{d\tau} = \frac{\rho - N\rho_1^*}{N D_N}, \quad \frac{d \log e_{BC}}{d\tau} = \frac{(N - 1)\rho}{N D_N} > 0. \quad (48)$$

At $N = 3$ these are (30); at $N = 2$ the first is (20), with ρ absent.

Result 7.1 (Dilution threshold). *A depreciates against each untariffed bystander iff*

$$\boxed{\rho > \rho_N^* = N\rho_1^*}. \quad (49)$$

Diverted expenditure is spread over $N - 2$ bystanders, each receiving weight $1/(N - 1)$, so increasingly strong cross-origin substitution is required to reverse any one bilateral rate. Dilution is a force one can expect in many models; the exact linearity in N is a property of the symmetric nested-CES structure. The threshold concerns the distribution of adjustment, not the aggregate value of A ’s currency:

$$\frac{d\nu_A^E}{d\tau} = \frac{\nu_B + (N - 2)\nu_C}{N - 1} = \boxed{-\frac{\rho_1^*}{D_N}} < 0, \quad \nu_B - \nu_C = -\frac{(N - 1)\rho}{N D_N} d\tau : \quad (50)$$

the ρ terms cancel exactly from the aggregate numerator for every $N \geq 2$, so N and ρ change the magnitude but not the sign of the symmetric NEER response, while the relative component is pure ρ .

Corollary (single elasticity). With $\rho = \eta = \sigma$,

$$\rho_N^* - \sigma = \sigma(N\alpha_D - 1) + N(1 - \alpha_D)(1 - \alpha_T) > 0 \quad \text{whenever } \alpha_D \geq 1/N : \quad (51)$$

any calibration with home share at least $1/N$ keeps the bilateral reversal invisible to single-elasticity models.

Correspondence across N .

object	$N = 2$	$N = 3$	general N
ML condition (43)	$\varepsilon_X + \varepsilon_M - 1 = D_2 > 0$	$D_3 > 0$	$-J$ a nonsingular M-matrix
sufficient primitive	$\eta \geq 1$	$\eta \geq 1$	gross substitutability (Assumption 2)
sign rule	one impact, unconditional	weights $\frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$	$\text{sign}(\sum_j \omega_{ij} F_j), \omega \geq 0$
bilateral threshold	none (no by-stander)	$3\rho_1^*$	$N\rho_1^*$ (symmetric); degree- $(N - 1)$ roots in general
real bilateral threshold	none	$\rho_q^*(3)$ (56)	$\rho_q^*(N)$ (57)
NEER (unilateral)	$-\rho_1^*/D_2$	$-\rho_1^*/D_3$	$-\rho_1^*/D_N$: sign invariant under symmetry
war threshold	—	ρ_1^*	ρ_1^* , independent of N

8 Alternative Tariff Configurations

Responses are linear in the tariff vector: configurations are sums of directed bilateral shocks (directions a applied to the same G), with mirror shocks obtained by relabeling (29).

8.1 Broad tariff

A uniform tariff on all partners treats every origin identically, so the cross-origin reallocation margin is inactive. Superposition gives, at $N = 3$ and general N ,

$$\frac{d \log e_{Aj}}{d\tau} = -\frac{2\rho_1^*}{D} \quad (N = 3), \quad \frac{d \log e_{Aj}}{d\tau} = -\frac{(N-1)\rho_1^*}{D_N} = \frac{d\nu_A^E}{d\tau} \quad \forall j : \quad (52)$$

ρ drops out of the numerator entirely, every bilateral rate appreciates equally, and bilateral and effective responses coincide. Bilateral ambiguity is generated by discrimination across foreign sup-

pliers, not by protection per se.

8.2 Symmetric trade war

B retaliates with an equal tariff on A . Adding the mirror shock:

$$\left. \frac{d \log e_{AB}}{d\tau} \right|^w = 0, \quad \left. \frac{d \log e_{AC}}{d\tau} \right|^w = \frac{\rho - \rho_1^*}{D_N}, \quad \left. \frac{d\nu_A^E}{d\tau} \right|^w = \frac{N-2}{N-1} \frac{\rho - \rho_1^*}{D_N}. \quad (53)$$

Result 8.1 (War threshold, independent of N). *In a symmetric war, every bystander appreciates against both belligerents, and both belligerents' NEERs depreciate, iff*

$$\boxed{\rho > \rho_1^* = \frac{\rho_N^*}{N}} \quad (54)$$

for any N . The unilateral tariff appreciates the NEER for all ρ ; the war depreciates it iff $\rho > \rho_1^*$.

Retaliation cancels the belligerents' bilateral component (e_{AB} is pinned at zero by symmetry), leaving the bystander diversion margin to determine their effective rates; each bystander's equation decouples, so its own impact effect decides the sign and does not dilute with N . The war threshold is far below the unilateral bilateral threshold $N\rho_1^*$ because the direct appreciation against B , which diversion must otherwise overcome, has been removed. At $\eta = 1$, $\rho_1^* = 1 - m < 1$, so any $\rho \geq 1$ suffices.

9 Real Exchange Rates

Real rates add the tariff's direct effect on A 's consumer prices. From (14) and (7), $d \log q_{Aj} = \nu_j + \alpha_T(d \log P_{T_j}^{\text{own}} - d \log P_{T_A}^{\text{own}})$; for the unilateral tariff at the symmetric baseline both bilateral real rates take one form at every N (verified symbolically):

$$d \log q_{Aj} = \left(1 - \frac{mN}{N-1}\right) \nu_j - \frac{m}{N-1} d\tau, \quad j = B, C, \quad (55)$$

i.e. $(1 - 2m)\nu_B - m d\tau$ at $N = 2$ and $(1 - \frac{3m}{2})\nu_j - \frac{m}{2} d\tau$ at $N = 3$: attenuated pass-through of the nominal rate plus a direct price-index term that works toward real appreciation of A at any nominal rate. Setting $d \log q_{AC} = 0$ with (48) gives the real bilateral thresholds

$$\rho_q^*(3) = \frac{(2 - 3m) 3\rho_1^* + 3m[3\alpha_D(\eta - 1) + 1]}{2(1 - 3m)} \quad (\rho_q^* > \rho_3^* \text{ for } m \in (0, \tfrac{1}{3})), \quad (56)$$

$$\boxed{\rho_q^*(N) = \frac{[(N-1) - mN] N\rho_1^* + mN[N\alpha_D(\eta - 1) + 1]}{(N-1)(1 - mN)}} \quad (m < 1/N), \quad (57)$$

with benchmark values $\rho_q^*(3) = 4.93$, $\rho_q^*(4) = 7.34$, $\rho_q^*(5) = 10.40$ against nominal thresholds 3.96, 5.28, 6.60. Averaging (55) gives the REER:

$$\frac{dq_A^E}{d\tau} = -\left(1 - \frac{mN}{N-1}\right) \frac{\rho_1^*}{D_N} - \frac{m}{N-1} < 0 \quad \text{for all } \rho, N \left(m < \frac{N-1}{N}\right). \quad (58)$$

Result 9.1 (Real rates). *Real bilateral reversal requires more diversion than nominal reversal, $\rho_q^*(N) > N\rho_1^*$, because part of the nominal movement is absorbed by the tariff's direct effect on A's price index. Real effective appreciation is unconditional and strictly stronger than nominal: the direct CPI effect reinforces the nominal effective appreciation.*

10 Asymmetry

The J - F framework does not require symmetry; the exact closed-form thresholds and the NEER cancellation do. Asymmetry reweights the same channels rather than introducing new ones. Two questions remain: where the bilateral threshold moves, and whether the aggregate result survives.

10.1 General asymmetric baseline

At any balanced baseline, (33)–(38) hold with observed shares (b_{ij}, h_i) and incomes y_i as coefficients. For $N = 3$ at a general baseline (unequal shares and sizes, existing tariffs; paper appendix), the bystander slope has denominator $\det(-J) > 0$ and numerator quadratic in ρ :

$$\text{sign} \frac{d \log e_{AC}}{d\tau} = \text{sign } \mathcal{N}(\rho), \quad \mathcal{N}(\rho) = c_2 \rho^2 + c_1 \rho + c_0, \quad c_2 = b_{AB} b_{AC} b_{CA} b_{CB} (1-h_A)(1-h_C) y_A y_C. \quad (59)$$

Since $c_2 > 0$ whenever A and C each buy from both foreign origins, sufficiently strong cross-origin substitution reverses e_{AC} at any such baseline: asymmetry moves the threshold (the largest root of $\mathcal{N}$, equal to $3\rho_1^*$ at the symmetric point) but cannot remove the reversal region. The coefficients c_1, c_0 reweight the symmetric channels: a country well positioned to replace B in A 's import basket receives more of the diversion effect, and a country heavily exposed to B 's income loss receives more of the offset, so no single universal threshold exists away from symmetry.

10.2 Structured asymmetry: the κ parameterization

To compare asymmetry across N , let the target carry a fixed multiple κ of the symmetric bilateral share (holding its absolute share fixed would not be comparable, since the symmetric share shrinks like $1/(N-1)$), with the weight matrix symmetric and hence balanced:

$$b_{AB} = b_{BA} = \frac{\kappa}{N-1}, \quad b_{AC} = b_{CA} = b_{BC} = b_{CB} = \frac{1 - \frac{\kappa}{N-1}}{N-2}, \quad b_{CC'} = \frac{1 - 2b_{CA}}{N-3}. \quad (60)$$

The bystanders remain interchangeable, so the reduced system again has two unknowns and solves in closed form (symbolically; replication package). With NEER weights $w_{Aj} = b_{Aj}$, the response

$d\nu_A^E/d\tau$ is an explicit rational function of (ρ, N, κ) that collapses to $-\rho_1^*/D_N$ exactly at $\kappa = 1$; its numerator is $-\kappa\rho_1^*$ times a quadratic $Q(\rho)$ with leading coefficient proportional to $(\kappa-1)(N-1-\kappa)$.

Result 10.1 (Asymmetric NEER). *Holding κ fixed as $N \rightarrow \infty$,*

$$N \frac{d\nu_A^E}{d\tau} \longrightarrow -\kappa \frac{\rho_1^*}{\rho + \alpha_D(\eta - 1)} = -\kappa \lim_{N \rightarrow \infty} \frac{N\rho_1^*}{D_N} : \quad (61)$$

to leading order, asymmetric exposure scales the symmetric NEER response by the target's relative importance and preserves its sign.

At finite N the sign follows from Q . For $\kappa \geq 1$ all coefficients of Q are positive at large N (the leading one by $(\kappa - 1)(N - 1 - \kappa) \geq 0$, the others with leading terms N^2 and $N^2\alpha_D(\eta - 1)$), so effective appreciation holds for every ρ . For $\kappa < 1$, Q has a single positive root

$$\rho_E^*(\kappa, N) = \frac{N\rho_1^*}{1 - \kappa} (1 + O(N^{-1})), \quad (62)$$

beyond which the NEER depreciates: a target with little NEER weight lets the depreciations against many bystanders outweigh the appreciation against B , and as $\kappa \rightarrow 0$ the flip point approaches the bilateral threshold $N\rho_1^*$. The reversal threshold is of order N ($\rho_E^* \approx 22$ at $N = 10$, $\kappa = 0.5$, benchmark), so the same dilution that protects bilateral rates protects the aggregate. The symmetric NEER result is exact under symmetry and approximately robust within this structured asymmetric family; robustness to arbitrary asymmetric trade networks is not claimed.

11 Relation to the Canonical PTA Model

11.1 Competing exporters

Countries A (importer), B and C (exporters); goods x and numeraire y ; A holds only y , B and C hold x . Quasilinear preferences $U_i = u_i(x_i) + y_i$ with $u'_i > 0$, $u''_i < 0$. A levies specific tariffs t_B, t_C ; with q_j the price received by exporter j and p the domestic price in A ,

$$p = q_B + t_B = q_C + t_C. \quad (63)$$

From $u'_i(x_i) = (\text{local price of } x)$, import demand and export supplies are

$$M_A(p) = x_A^d(p), \quad E_j(q_j) = \omega_j - x_j^d(q_j), \quad j = B, C, \quad M'_A < 0, \quad E'_j > 0, \quad (64)$$

and market clearing for x determines p given (t_B, t_C) :

$$M_A(p) = E_B(p - t_B) + E_C(p - t_C). \quad (65)$$

Differentiating in t_B at fixed t_C :

$$\frac{dp}{dt_B} = \frac{E'_B}{E'_B + E'_C - M'_A} \in (0, 1) \implies \frac{dq_B}{dt_B} = -\frac{E'_C - M'_A}{E'_B + E'_C - M'_A} < 0, \quad \frac{dq_C}{dt_B} = \frac{dp}{dt_B} > 0. \quad (66)$$

Result 11.1 (Viner–Mundell). *A discriminatory tariff on B worsens B’s terms of trade and improves C’s terms of trade, unconditionally.*

11.2 Which margins are missing?

An improvement in C ’s terms of trade is a rise in e_{AC} (13); the nested model requires $\rho > 3\rho_1^*$ for it, while the canonical model signs it unconditionally. The difference is the set of margins permitted. Decompose the threshold (benchmark values):

$$\rho_3^* = 3\rho_1^* = \underbrace{3}_{B-C \text{ linkage}} + \underbrace{3\alpha_D(\eta - 1)}_{\text{home switching}} - \underbrace{3\alpha_T(1 - \alpha_D)}_{\text{openness credit}} = 3.0 + 1.2 - 0.24 = 3.96. \quad (67)$$

The first term collects the B – C trade linkage: B ’s income loss cuts its purchases from C , and relative prices induce C to substitute toward B ’s cheapened goods. The second is home switching in A , strengthening with η and α_D . The third credits openness: a larger import share m gives the tariff more foreign expenditure to reallocate. Recomputing the threshold with each margin removed confirms the decomposition (severing trade changes the baseline, so the pieces are approximately additive):

restriction	threshold ρ_3^*	piece removed
none (benchmark)	3.96	—
no B – C trade ($\beta_{BC} = \beta_{CB} = 0$)	1.05	the linkage
no home tradables ($\alpha_D = 0$)	1.80	home switching
no non-tradables ($\alpha_T = 1$)	3.60	credit maximized
$\alpha_D = 0$ and $\alpha_T = 1$	none: reversal for all $\rho > 0$	
no B – C trade and $\alpha_D = 0$	none: reversal for all $\rho > 0$	

The largest piece is the linkage; two routes reach $\rho_3^* \leq 0$, and the canonical model takes both. It makes B and C competing exporters rather than full trading partners (no linkage), lets A produce none of the imported good (no home switching), and imposes quasilinearity (no income effects); the Viner case adds $\rho = \infty$. What remains is reallocation between the two foreign suppliers, signed directly by $M'_A < 0$ and $E'_j > 0$.

Result 11.2. *The canonical PTA model is the restriction of the three-country model to a subspace of structures on which $\rho_3^* \leq 0$: the unconditional Viner–Mundell sign is the threshold condition evaluated where it cannot bind. The nested model does not overturn the Viner–Mundell mechanism; it embeds it among the opposing margins, and $\rho > \rho_3^*$ states when diversion dominates them.*

The analytical PTA literature obtains sharp signs by such restrictions (competing exporters: Bagwell–Staiger 1999, Ornelas 2005; symmetric endowment blocs: Krugman 1991, Bond–Syropoulos 1996; Viner: $\rho = \infty$; Kemp–Wan 1976: no restrictions and no sign claims). Applied models in the GTAP and quantitative-trade traditions retain every margin in (67) and generate both signs in simulations, but state no condition separating them; the threshold is that condition. The threshold is not an artifact of CES curvature: each of its pieces is an identifiable economic force absent from the canonical setup.

12 Conclusion

Two countries: under the Marshall–Lerner condition, a unilateral import tariff appreciates the tariffing country’s currency; the tariff signs the disturbance and Marshall–Lerner signs the adjustment.

Three countries: foreign-supplier reallocation makes bilateral responses ambiguous. A always appreciates against the target, but depreciates against an untreated country iff $\rho > 3\rho_1^*$. In general equilibrium the tariff creates a disturbance vector F and the adjustment mechanism $(-J)^{-1} \geq 0$ combines it: mixed bilateral signs require a mixed F , never a perverse adjustment mechanism.

Under symmetry, aggregation removes the direct foreign-reallocation component: the NEER appreciates for every ρ and every N , at rate ρ_1^*/D_N . The conventional two-country result survives multilateralization as an effective, not universally bilateral, statement. At general N , diversion toward each bystander is diluted and the bilateral threshold rises to $N\rho_1^*$, while the NEER response is unchanged in sign.

The limits of the aggregate result are set by configuration and asymmetry. A broad tariff deactivates the reallocation margin entirely; a symmetric war cancels the belligerents’ bilateral component and depreciates their effective rates iff $\rho > \rho_1^*$, independently of N . Under asymmetry the exact cancellation is special: within the κ -parameterized family the aggregate sign is preserved to leading order and for all over-exposed targets, but a sufficiently under-weighted target combined with ρ of order N can reverse it.

The organizing distinction throughout is between aggregate home-versus-foreign adjustment, which determines the sign of the effective exchange-rate response, and reallocation across foreign suppliers, which distributes the adjustment across bilateral rates and, through equilibrium, affects the magnitude of the aggregate response. Tariffs change both how much a country buys from abroad and from whom it buys it; the first margin governs the currency’s aggregate response, the second its bilateral incidence.